Lecture 11: Node Voltage Analysis

EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current voltage, node voltage e supernode

Review: two ways to solve an LTI resistive circuit

Transform the circuit

- Substitution, equivalent one-ports.
- Series and parallel connections.
- Source transformation.

Works for many circuits, but not all.

Write its equations as it is

- KCL: $A\mathbf{i} = \mathbf{0}$.
- KVL: $\mathbf{v} = A^T\mathbf{e}$.
- One element law per branch.

Works for every circuit.

With the node voltages $\mathbf{e}$ as the unknowns, KVL holds automatically.

Only KCL is left to write.

Why this lecture?

- **So far:** we simplified circuits by transforming them.

Why this lecture?

- **So far:** we simplified circuits by transforming them.
- **Now:** we write the equations of the circuit as it is, with a systematic procedure: the same steps for every circuit.

Why this lecture?

- **So far:** we simplified circuits by transforming them.
- **Now:** we write the equations of the circuit as it is, with a systematic procedure: the same steps for every circuit.
- **The unknowns:** only the $n-1$ node voltages, not all $2b$ branch voltages and currents.

Why this lecture?

- **So far:** we simplified circuits by transforming them.
- **Now:** we write the equations of the circuit as it is, with a systematic procedure: the same steps for every circuit.
- **The unknowns:** only the $n-1$ node voltages, not all $2b$ branch voltages and currents.
- **The result:** a linear system that works for every LTI resistive circuit.

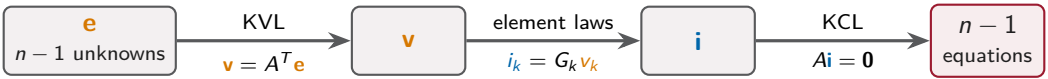
Why this lecture?

- **So far:** we simplified circuits by transforming them.
- **Now:** we write the equations of the circuit as it is, with a systematic procedure: the same steps for every circuit.
- **The unknowns:** only the $n-1$ node voltages, not all $2b$ branch voltages and currents.
- **The result:** a linear system that works for every LTI resistive circuit.
- **Later:** mesh current analysis, the same idea with KCL and KVL swapped.

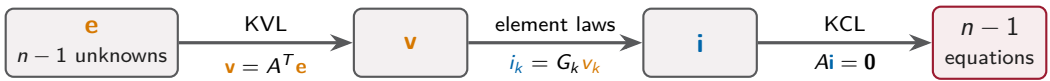
Roadmap

1. **The idea** — node voltages as the unknowns
2. **Only current sources** — $Y_n \mathbf{e} = \mathbf{i}_s$, written by inspection
3. **A voltage source to the ground** — a node voltage is known
4. **A voltage source between two nodes** — Method I: a new unknown current
5. **The supernode** — Method II: KCL around both nodes
6. **Dependent sources and the ground** — constraint equations, another ground

The idea: node voltages as the unknowns

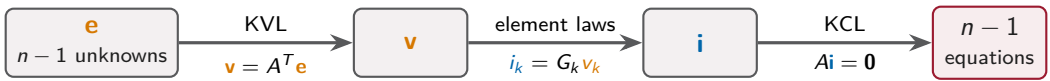


The idea: node voltages as the unknowns



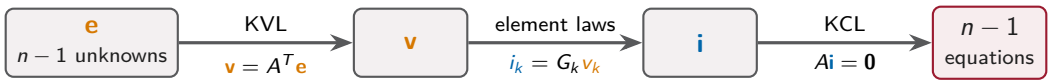
- Write KCL at each node except the reference node, with every current in terms of node voltages.

The idea: node voltages as the unknowns



- Write KCL at each node except the reference node, with every current in terms of node voltages.
- This gives $n-1$ equations in $n-1$ unknowns.

The idea: node voltages as the unknowns



- Write KCL at each node except the reference node, with every current in terms of node voltages.
- This gives $n-1$ equations in $n-1$ unknowns.
- Once $\mathbf{e}$ is known, $\mathbf{v} = A^T\mathbf{e}$ gives the branch voltages, and the element laws give the branch currents.

The procedure

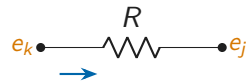
1. Choose a reference node (ground); its node voltage is 0.

The procedure

1. Choose a reference node (ground); its node voltage is 0.
2. Label the other node voltages $e_1, \ldots, e_{n-1}$.

The procedure

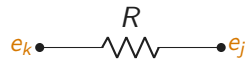
1. Choose a reference node (ground); its node voltage is 0.
2. Label the other node voltages $e_1, \ldots, e_{n-1}$.
3. Write KCL at every other node, in terms of node voltages.



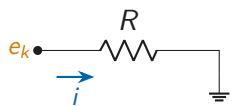
$$i = \frac{e_k - e_j}{R} = G\,(e_k - e_j)$$

The procedure

1. Choose a reference node (ground); its node voltage is 0.
2. Label the other node voltages $e_1, \ldots, e_{n-1}$.
3. Write KCL at every other node, in terms of node voltages.



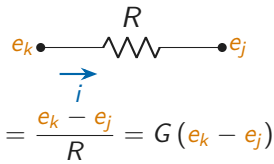
$$i = \frac{e_k - e_j}{R} = G\,(e_k - e_j)$$

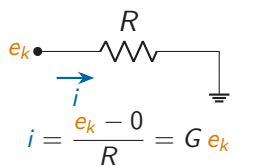


$$i = \frac{e_k - 0}{R} = G\,e_k$$

The procedure

1. Choose a reference node (ground); its node voltage is 0.
2. Label the other node voltages $e_1, \ldots, e_{n-1}$.
3. Write KCL at every other node, in terms of node voltages.



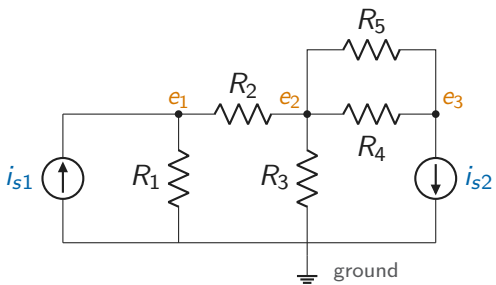


The current leaving node k through a resistor is its own node voltage minus the other one, over R.

Roadmap

1. **The idea** — node voltages as the unknowns ✓
2. **Only current sources** — $Y_n \mathbf{e} = \mathbf{i}_s$, written by inspection
3. **A voltage source to the ground** — a node voltage is known
4. **A voltage source between two nodes** — Method I: a new unknown current
5. **The supernode** — Method II: KCL around both nodes
6. **Dependent sources and the ground** — constraint equations, another ground

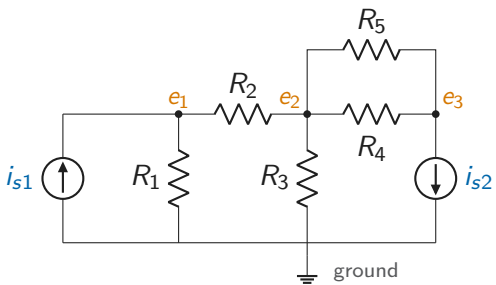
Example 11.1: only current sources



Question. How many unknowns does this circuit have?

How many does node voltage analysis need?

Example 11.1: only current sources

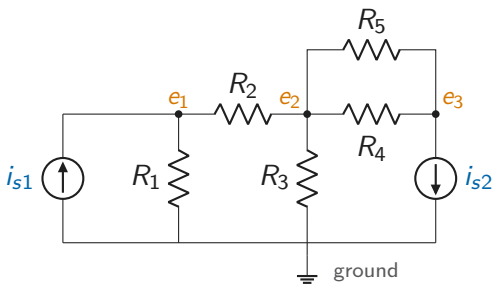


Question. How many unknowns does this circuit have?

How many does node voltage analysis need?

$b = 7$ branches: $2b = 14$ branch voltages and currents.

Example 11.1: only current sources



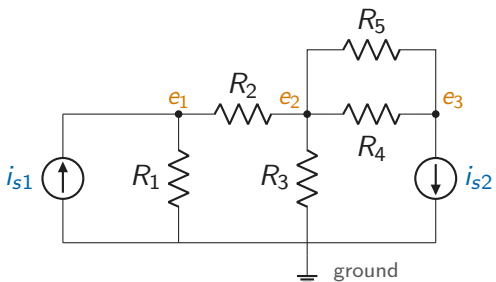
Question. How many unknowns does this circuit have?

How many does node voltage analysis need?

$b = 7$ branches: $2b = 14$ branch voltages and currents.

$n = 4$ nodes: only 3 node voltages, e_1, e_2, e_3.

Example 11.1: only current sources



Question. How many unknowns does this circuit have?

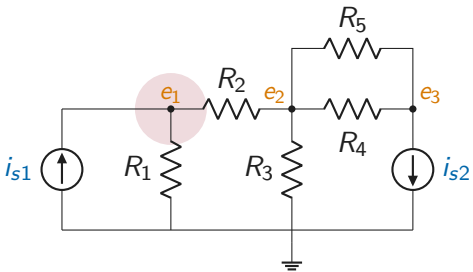
How many does node voltage analysis need?

$b = 7$ branches: $2b = 14$ branch voltages and currents.

$n = 4$ nodes: only 3 node voltages, e_1, e_2, e_3.

So we need three KCL equations, at nodes 1, 2 and 3.

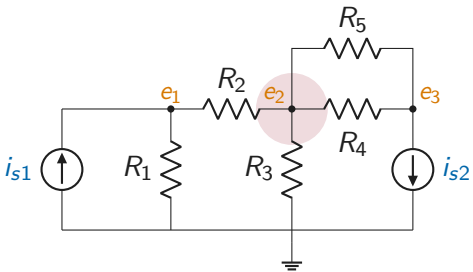
Example 11.1: KCL at each node



Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

Example 11.1: KCL at each node



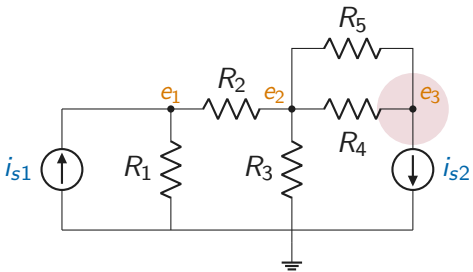
Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

Node 2:

$$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} + \frac{e_2 - e_3}{R_5} = 0$$

Example 11.1: KCL at each node



Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

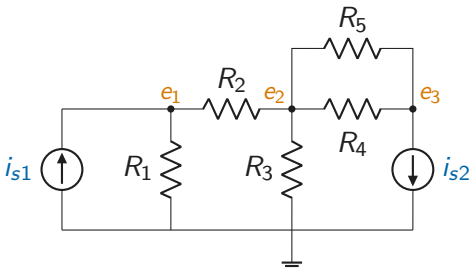
Node 2:

$$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} + \frac{e_2 - e_3}{R_5} = 0$$

Node 3:

$$\frac{e_3 - e_2}{R_4} + \frac{e_3 - e_2}{R_5} + i_{s2} = 0$$

Example 11.1: KCL at each node



Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

Node 2:

$$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} + \frac{e_2 - e_3}{R_5} = 0$$

Node 3:

$$\frac{e_3 - e_2}{R_4} + \frac{e_3 - e_2}{R_5} + i_{s2} = 0$$

Every term is a current leaving the node.

A source pointing into the node enters with a minus sign.

Example 11.1: matrix form

With the conductances $G_k = 1/R_k$:

$$\begin{aligned}
(G_1 + G_2)\, e_1 - G_2\, e_2 &= i_{s1} \\
-G_2\, e_1 + (G_2 + G_3 + G_4 + G_5)\, e_2 - (G_4 + G_5)\, e_3 &= 0 \\
-(G_4 + G_5)\, e_2 + (G_4 + G_5)\, e_3 &= -i_{s2}
\end{aligned}$$

Example 11.1: matrix form

With the conductances $G_k = 1/R_k$:

$$
\begin{aligned}
(G_1 + G_2)\, e_1 - G_2\, e_2 &= i_{s1} \\
-G_2\, e_1 + (G_2 + G_3 + G_4 + G_5)\, e_2 - (G_4 + G_5)\, e_3 &= 0 \\
-(G_4 + G_5)\, e_2 + (G_4 + G_5)\, e_3 &= -i_{s2}
\end{aligned}
$$

$$
\underbrace{\begin{bmatrix} G_1 + G_2 & -G_2 & 0 \\ -G_2 & G_2 + G_3 + G_4 + G_5 & -(G_4 + G_5) \\ 0 & -(G_4 + G_5) & G_4 + G_5 \end{bmatrix}}_{Y_n}
\underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}}
=
\underbrace{\begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix}}_{\mathbf{i}_s}
$$

Example 11.1: matrix form

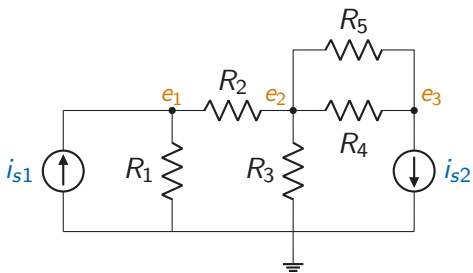
With the conductances $G_k = 1/R_k$:

$$\begin{aligned}
(G_1 + G_2)\, e_1 - G_2\, e_2 &= i_{s1} \\
-G_2\, e_1 + (G_2 + G_3 + G_4 + G_5)\, e_2 - (G_4 + G_5)\, e_3 &= 0 \\
-(G_4 + G_5)\, e_2 + (G_4 + G_5)\, e_3 &= -i_{s2}
\end{aligned}$$

$$\underbrace{\begin{bmatrix} G_1 + G_2 & -G_2 & 0 \\ -G_2 & G_2 + G_3 + G_4 + G_5 & -(G_4 + G_5) \\ 0 & -(G_4 + G_5) & G_4 + G_5 \end{bmatrix}}_{Y_n} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}} = \underbrace{\begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix}}_{\mathbf{i}_s}$$

Y_n is the node admittance matrix: $Y_n \mathbf{e} = \mathbf{i}_s$.

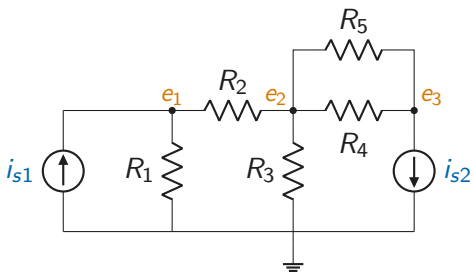
Writing Y_n by inspection



$$\begin{bmatrix} G_1 + G_2 & -G_2 & 0 \\ -G_2 & G_2 + \cdots + G_5 & -(G_4 + G_5) \\ 0 & -(G_4 + G_5) & G_4 + G_5 \end{bmatrix}$$

Question. Can you read each entry of Y_n directly from the circuit?

Writing Y_n by inspection

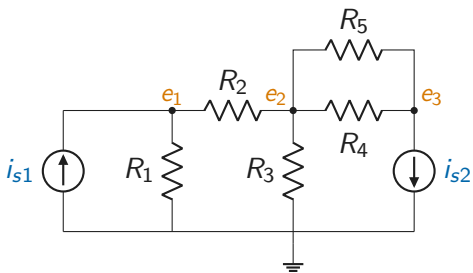


$$\begin{bmatrix} G_1 + G_2 & -G_2 & 0 \\ -G_2 & G_2 + \cdots + G_5 & -(G_4 + G_5) \\ 0 & -(G_4 + G_5) & G_4 + G_5 \end{bmatrix}$$

Question. Can you read each entry of Y_n directly from the circuit?

Diagonal (k, k): the sum of the conductances connected to node k.

Writing Y_n by inspection



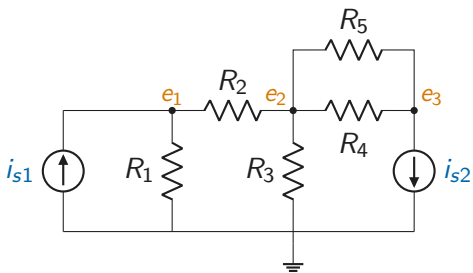
$$\begin{bmatrix} G_1+G_2 & -G_2 & 0 \\ -G_2 & G_2+\cdots+G_5 & -(G_4+G_5) \\ 0 & -(G_4+G_5) & G_4+G_5 \end{bmatrix}$$

Question. Can you read each entry of Y_n directly from the circuit?

Diagonal (k,k): the sum of the conductances connected to node k.

Off-diagonal (k,j): minus the sum of the conductances between nodes k and j.

Writing Y_n by inspection



$$\begin{bmatrix} G_1+G_2 & -G_2 & 0 \\ -G_2 & G_2+\cdots+G_5 & -(G_4+G_5) \\ 0 & -(G_4+G_5) & G_4+G_5 \end{bmatrix}$$

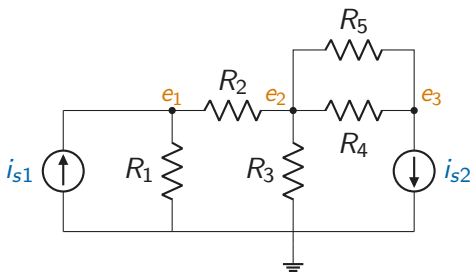
Question. Can you read each entry of Y_n directly from the circuit?

Diagonal (k,k): the sum of the conductances connected to node k.

Off-diagonal (k,j): minus the sum of the conductances between nodes k and j.

Right side k: the sum of the source currents entering node k.

Writing Y_n by inspection



$$\begin{bmatrix} G_1+G_2 & -G_2 & 0 \\ -G_2 & G_2+\cdots+G_5 & -(G_4+G_5) \\ 0 & -(G_4+G_5) & G_4+G_5 \end{bmatrix}$$

Question. Can you read each entry of Y_n directly from the circuit?

Diagonal (k,k): the sum of the conductances connected to node k.

Off-diagonal (k,j): minus the sum of the conductances between nodes k and j.

Right side k: the sum of the source currents entering node k.

So Y_n is symmetric: the entry (k,j) equals the entry (j,k).

Why is Y_n symmetric?

Split A into the resistor branches and the current source branches: $A = [\, A_R \;\; A_S \,]$.

Why is Y_n symmetric?

Split A into the resistor branches and the current source branches: $A = [\, A_R \;\; A_S \,]$.

- KCL: $A_R \mathbf{i}_R + A_S \mathbf{i}_S = \mathbf{0}$.

Why is Y_n symmetric?

Split A into the resistor branches and the current source branches: $A = [\, A_R \;\; A_S \,]$.

- KCL: $A_R \mathbf{i}_R + A_S \mathbf{i}_S = \mathbf{0}$.
- Resistors: $\mathbf{i}_R = G\,\mathbf{v}_R$,
  $G = \mathrm{diag}(G_1, \ldots, G_5)$.

Why is Y_n symmetric?

Split A into the resistor branches and the current source branches: $A = [\, A_R \;\; A_S \,]$.

- KCL: $A_R \mathbf{i}_R + A_S \mathbf{i}_S = \mathbf{0}$.
- Resistors: $\mathbf{i}_R = G\,\mathbf{v}_R$,
  $G = \mathrm{diag}(G_1, \ldots, G_5)$.
- KVL: $\mathbf{v}_R = A_R^T \mathbf{e}$.

Why is Y_n symmetric?

Split A into the resistor branches and the current source branches: $A = [\, A_R \;\; A_S \,]$.

- KCL: $A_R \mathbf{i}_R + A_S \mathbf{i}_S = \mathbf{0}$.
- Resistors: $\mathbf{i}_R = G\,\mathbf{v}_R$, $G = \mathrm{diag}(G_1, \ldots, G_5)$.
- KVL: $\mathbf{v}_R = A_R^T \mathbf{e}$.

$$\underbrace{A_R\, G\, A_R^T}_{Y_n}\, \mathbf{e} = \underbrace{-A_S \mathbf{i}_S}_{\mathbf{i}_s}$$

Why is Y_n symmetric?

Split A into the resistor branches and the current source branches: $A = [\, A_R \;\; A_S \,]$.

- KCL: $A_R \mathbf{i}_R + A_S \mathbf{i}_S = \mathbf{0}$.
- Resistors: $\mathbf{i}_R = G\,\mathbf{v}_R$, $G = \mathrm{diag}(G_1, \ldots, G_5)$.
- KVL: $\mathbf{v}_R = A_R^T \mathbf{e}$.

$$\underbrace{A_R\, G\, A_R^T}_{Y_n}\, \mathbf{e} = \underbrace{-A_S \mathbf{i}_S}_{\mathbf{i}_s}$$

$$Y_n^T = A_R\, G^T A_R^T = A_R\, G\, A_R^T = Y_n$$

Why is Y_n symmetric?

Split A into the resistor branches and the current source branches: $A = [\, A_R \;\; A_S \,]$.

- KCL: $A_R \mathbf{i}_R + A_S \mathbf{i}_S = \mathbf{0}$.
- Resistors: $\mathbf{i}_R = G\,\mathbf{v}_R$, $G = \mathrm{diag}(G_1, \ldots, G_5)$.
- KVL: $\mathbf{v}_R = A_R^T \mathbf{e}$.

$$\underbrace{A_R\, G\, A_R^T}_{Y_n}\, \mathbf{e} = \underbrace{-A_S \mathbf{i}_S}_{\mathbf{i}_s}$$

$$Y_n^T = A_R\, G^T A_R^T = A_R\, G\, A_R^T = Y_n$$

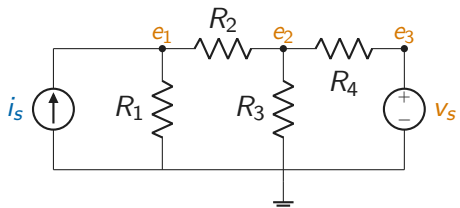
G is diagonal, so Y_n is symmetric.

$-A_S \mathbf{i}_S$ adds up the source currents entering each node.

Roadmap

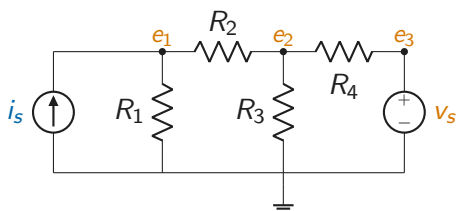
1. **The idea** — node voltages as the unknowns ✓
2. **Only current sources** — $Y_n \mathbf{e} = \mathbf{i}_s$, written by inspection ✓
3. **A voltage source to the ground** — a node voltage is known
4. **A voltage source between two nodes** — Method I: a new unknown current
5. **The supernode** — Method II: KCL around both nodes
6. **Dependent sources and the ground** — constraint equations, another ground

Example 11.2: a voltage source to the ground



Unknowns e_1, e_2, e_3: three equations needed.

Example 11.2: a voltage source to the ground

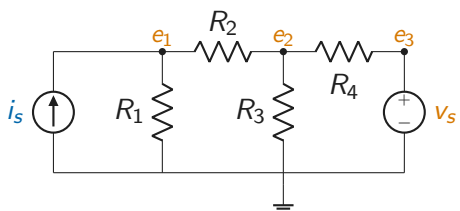


Unknowns e_1, e_2, e_3: three equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Example 11.2: a voltage source to the ground



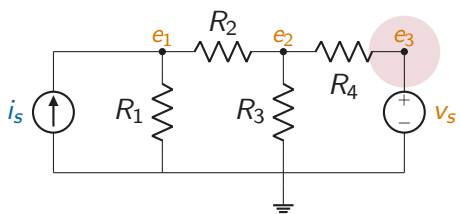
Unknowns e_1, e_2, e_3: three equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Question. What does KCL at node 3 give?

Example 11.2: a voltage source to the ground



Unknowns e_1, e_2, e_3: three equations needed.

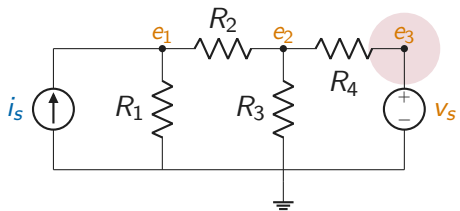
Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Question. What does KCL at node 3 give?

The current through v_s is unknown: KCL gives no useful equation in e_1, e_2, e_3.

Example 11.2: a voltage source to the ground



Unknowns e_1, e_2, e_3: three equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Question. What does KCL at node 3 give?

The current through v_s is unknown: KCL gives no useful equation in e_1, e_2, e_3.

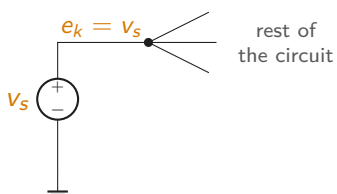
But $e_3 = v_s$: this is the third equation.

A voltage source to the ground

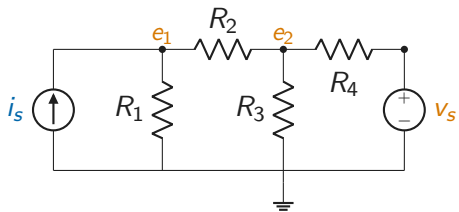
Remark

If an independent voltage source v_s is connected between the ground and node k, then $e_k = v_s$ is known.

So e_k is not an unknown, and we do not write KCL at node k.



Example 11.2, revisited

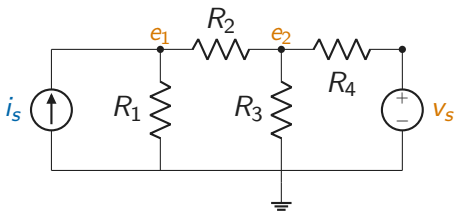


Unknowns e_1, e_2 only: two equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

Example 11.2, revisited



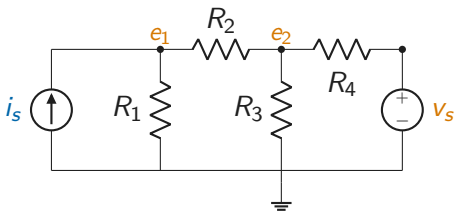
Unknowns e_1, e_2 only: two equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

Example 11.2, revisited



Unknowns e_1, e_2 only: two equations needed.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

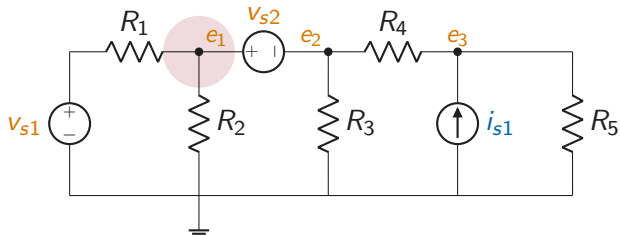
The known v_s moves to the right side.

v_s/R_4 acts like a current source into node 2: a source transformation.

Roadmap

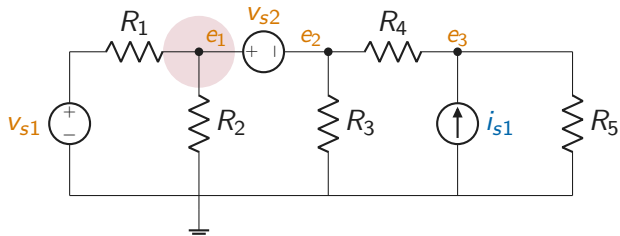
1. **The idea** — node voltages as the unknowns ✓
2. **Only current sources** — $Y_n \mathbf{e} = \mathbf{i}_s$, written by inspection ✓
3. **A voltage source to the ground** — a node voltage is known ✓
4. **A voltage source between two nodes** — Method I: a new unknown current
5. **The supernode** — Method II: KCL around both nodes
6. **Dependent sources and the ground** — constraint equations, another ground

Example 11.3: a voltage source between two nodes



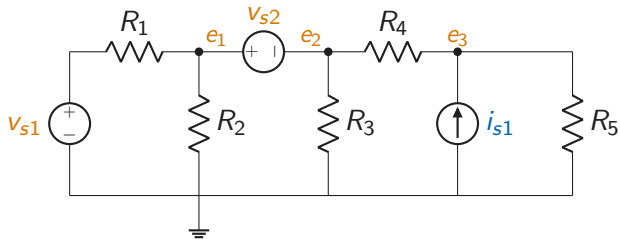
- **Question.** Write KCL at node 1: what is the current through v_{s2}?

Example 11.3: a voltage source between two nodes



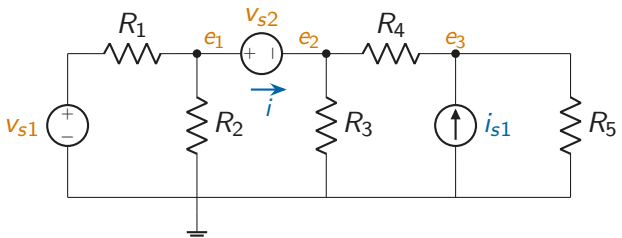
- **Question.** Write KCL at node 1: what is the current through v_{s2}?
- It cannot be written with e_1 and e_2: the law of a voltage source fixes its voltage, not its current.

Example 11.3: a voltage source between two nodes



- **Question.** Write KCL at node 1: what is the current through v_{s2}?
- It cannot be written with e_1 and e_2: the law of a voltage source fixes its voltage, not its current.
- **Modified node voltage analysis** offers two ways out: Method I (its current is a new unknown) and Method II (a supernode).

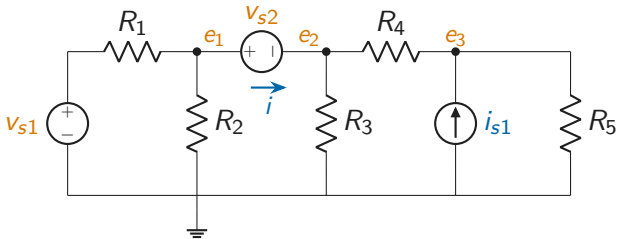
Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Method I: the current as an unknown

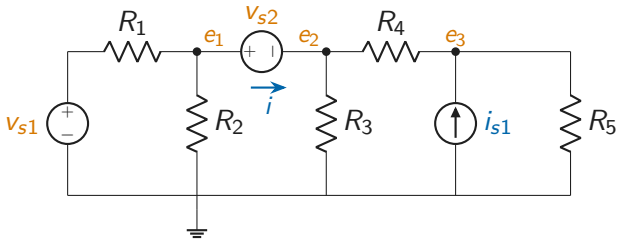


Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Method I: the current as an unknown



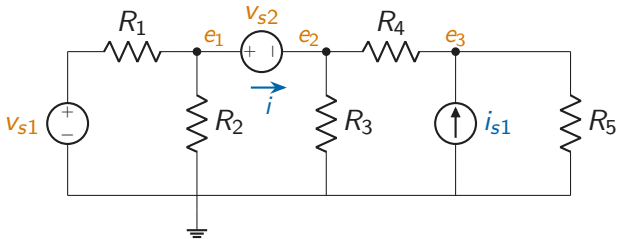
Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

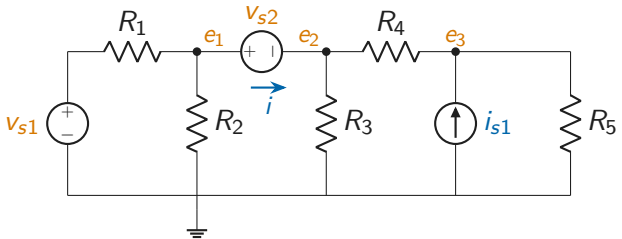
Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Four unknowns, three equations: one more is needed.

Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Unknowns: e_1, e_2, e_3 and the current i.

Four unknowns, three equations: one more is needed.

Constraint: $e_1 - e_2 = v_{s2}$, the law of the source.

Method I: matrix form

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

Method I: matrix form

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

- The first three rows are KCL; the last row is the constraint.

Method I: matrix form

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

- The first three rows are KCL; the last row is the constraint.
- The unknown current i is part of the solution: we get it directly.

Method I: matrix form

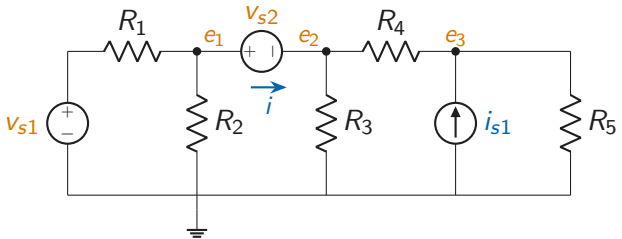
$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

- The first three rows are KCL; the last row is the constraint.
- The unknown current i is part of the solution: we get it directly.
- The price: one more unknown and one more equation.

Roadmap

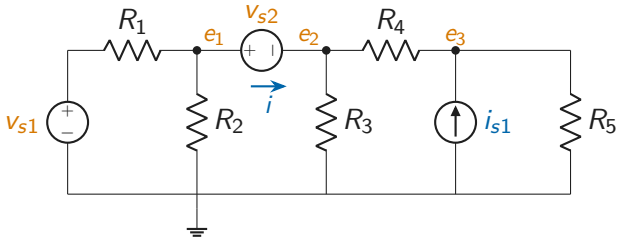
1. **The idea** — node voltages as the unknowns ✓
2. **Only current sources** — $Y_n \mathbf{e} = \mathbf{i}_s$, written by inspection ✓
3. **A voltage source to the ground** — a node voltage is known ✓
4. **A voltage source between two nodes** — Method I: a new unknown current ✓
5. **The supernode** — Method II: KCL around both nodes
6. **Dependent sources and the ground** — constraint equations, another ground

Method II: the supernode



Question. Can we get rid of the unknown i?

Method II: the supernode

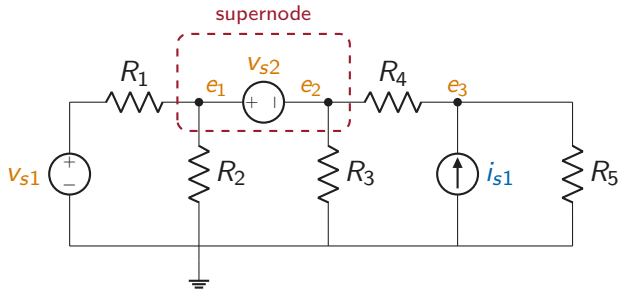


Question. Can we get rid of the unknown i?

Add the KCL equations of nodes 1 and 2.

i leaves node 1 and enters node 2, so it cancels.

Method II: the supernode



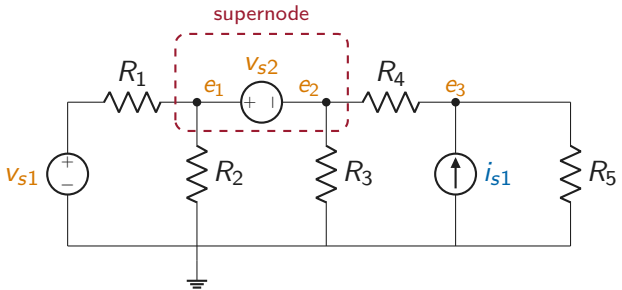
Question. Can we get rid of the unknown i?

Add the KCL equations of nodes 1 and 2.

i leaves node 1 and enters node 2, so it cancels.

This is KCL for a closed surface around both nodes and the source.

Method II: the supernode



Question. Can we get rid of the unknown i?

Add the KCL equations of nodes 1 and 2.

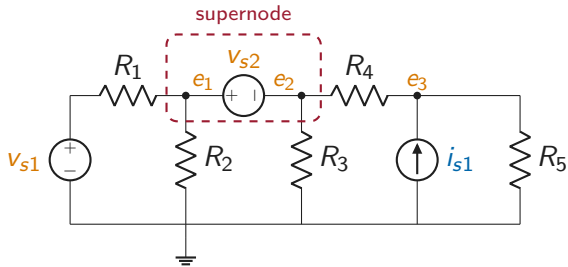
i leaves node 1 and enters node 2, so it cancels.

This is KCL for a closed surface around both nodes and the source.

Supernode

Write KCL as if the two nodes of the voltage source were a single node.

Method II: the equations



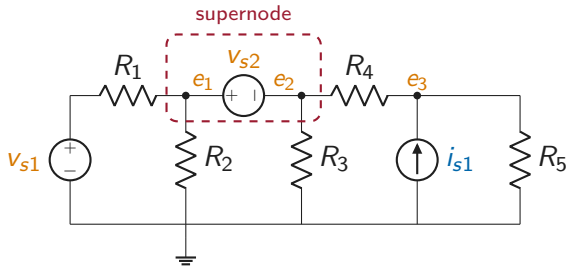
Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

Method II: the equations



Supernode:

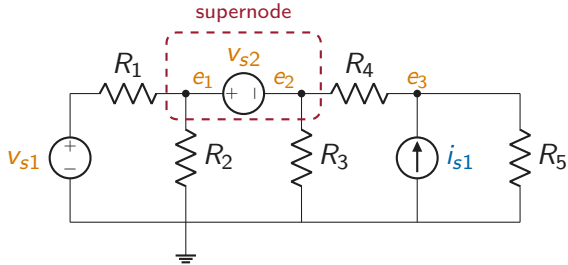
$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

Method II: the equations



Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

Three unknowns, three equations.

The matrix is not symmetric any more.

Method I or Method II?

	Method I	**Method II**
extra unknown	the current i of the source	none
KCL	at every non-ground node	at the supernode, and the others
extra equation	the constraint $e_1 - e_2 = v_{s2}$	the same constraint
size (this example)	4×4	3×3
the current i	part of the solution	from KCL afterwards

Method I or Method II?

	Method I	**Method II**
extra unknown	the current i of the source	none
KCL	at every non-ground node	at the supernode, and the others
extra equation	the constraint $e_1 - e_2 = v_{s2}$	the same constraint
size (this example)	4×4	3×3
the current i	part of the solution	from KCL afterwards

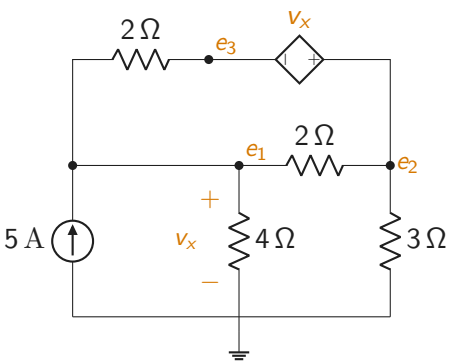
Both give the same node voltages.

The constraint is always the law of the voltage source.

Roadmap

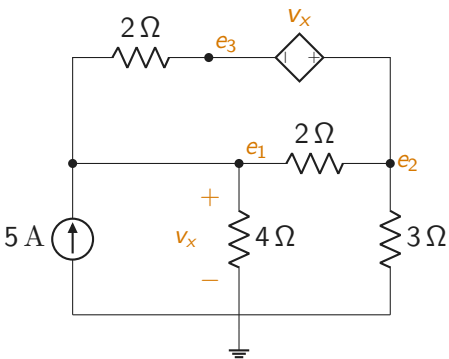
1. **The idea** — node voltages as the unknowns ✓
2. **Only current sources** — $Y_n \mathbf{e} = \mathbf{i}_s$, written by inspection ✓
3. **A voltage source to the ground** — a node voltage is known ✓
4. **A voltage source between two nodes** — Method I: a new unknown current ✓
5. **The supernode** — Method II: KCL around both nodes ✓
6. **Dependent sources and the ground** — constraint equations, another ground

Example 11.4: a dependent source



Question. Find the power absorbed by the 3 Ω resistor.

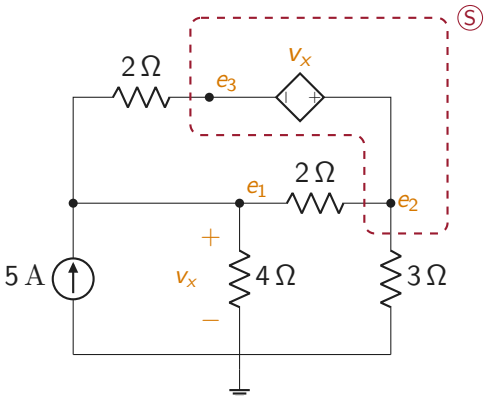
Example 11.4: a dependent source



Question. Find the power absorbed by the 3 Ω resistor.

The voltage source v_x is between nodes 3 and 2; neither is the ground.

Example 11.4: a dependent source

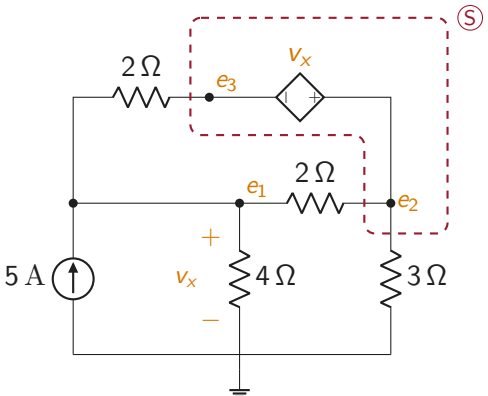


Question. Find the power absorbed by the 3 Ω resistor.

The voltage source v_x is between nodes 3 and 2; neither is the ground.

So use a supernode (S) around nodes 2 and 3.

Example 11.4: a dependent source



Question. Find the power absorbed by the 3 Ω resistor.

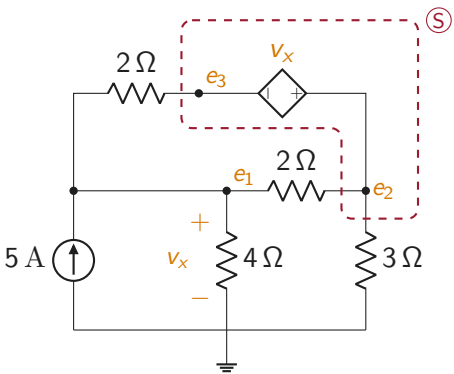
The voltage source v_x is between nodes 3 and 2; neither is the ground.

So use a supernode Ⓢ around nodes 2 and 3.

Unknowns: e_1, e_2, e_3.

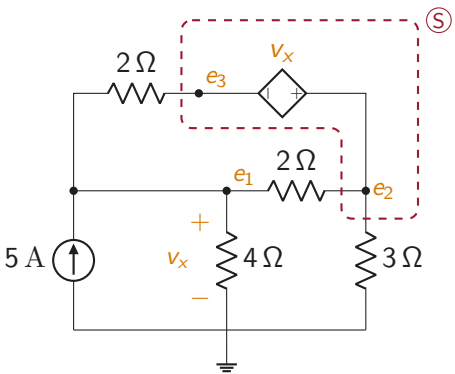
Equations: KCL at node 1, KCL at Ⓢ, and a constraint.

Example 11.4: the equations



Node 1: $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

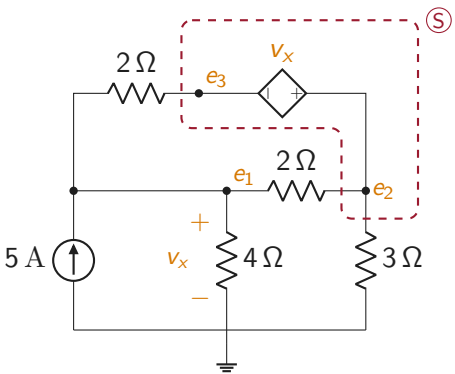
Example 11.4: the equations



Node 1: $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

Supernode Ⓢ: $$\frac{e_3 - e_1}{2} + \frac{e_2 - e_1}{2} + \frac{e_2}{3} = 0 \quad (2)$$

Example 11.4: the equations

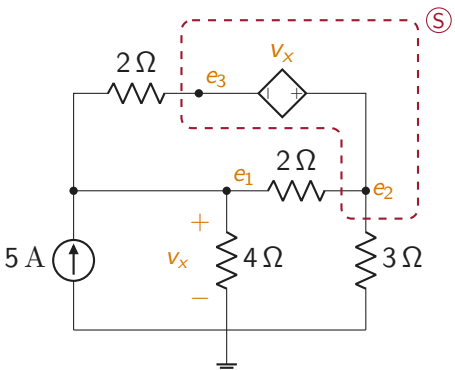


Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0 \quad (1)$

Supernode Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0 \quad (2)$

Question. What is the constraint?

Example 11.4: the equations



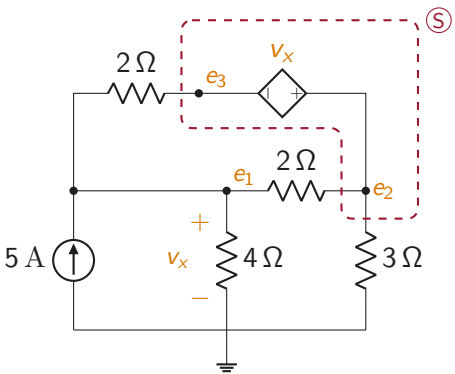
Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

Supernode Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

Question. What is the constraint?

The source: $e_2 - e_3 = v_x$.

Example 11.4: the equations



Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

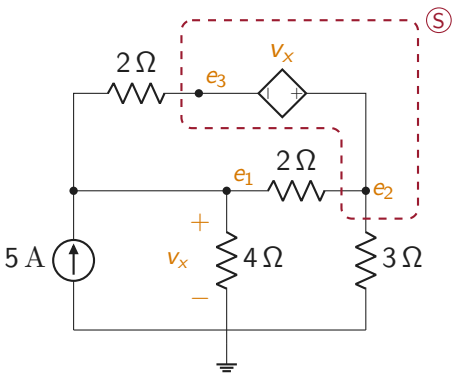
Supernode Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

Question. What is the constraint?

The source: $e_2 - e_3 = v_x$.

Its control: $v_x = e_1$, the voltage of the 4 Ω.

Example 11.4: the equations



Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

Supernode Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

Question. What is the constraint?

The source: $e_2 - e_3 = v_x$.

Its control: $v_x = e_1$, the voltage of the 4 Ω.

Constraint: $e_2 - e_3 = e_1$ (3)

Three equations, three unknowns.

Example 11.4: solve

- (1): $\frac{5}{4}e_1 - \frac{1}{2}e_2 - \frac{1}{2}e_3 = 5$; put $e_1 = e_2 - e_3$ from (3): $\frac{3}{4}e_2 - \frac{7}{4}e_3 = 5$ (4)

Example 11.4: solve

- (1): $\frac{5}{4}e_1 - \frac{1}{2}e_2 - \frac{1}{2}e_3 = 5$; put $e_1 = e_2 - e_3$ from (3): $\frac{3}{4}e_2 - \frac{7}{4}e_3 = 5$ (4)
- (2): $-e_1 + \frac{5}{6}e_2 + \frac{1}{2}e_3 = 0$; put (3): $-\frac{1}{6}e_2 + \frac{3}{2}e_3 = 0 \Rightarrow e_3 = \frac{1}{9}e_2$ (5)

Example 11.4: solve

- (1): $\frac{5}{4}e_1 - \frac{1}{2}e_2 - \frac{1}{2}e_3 = 5$; put $e_1 = e_2 - e_3$ from (3): $\frac{3}{4}e_2 - \frac{7}{4}e_3 = 5$ (4)
- (2): $-e_1 + \frac{5}{6}e_2 + \frac{1}{2}e_3 = 0$; put (3): $-\frac{1}{6}e_2 + \frac{3}{2}e_3 = 0 \Rightarrow e_3 = \frac{1}{9}e_2$ (5)
- (4) and (5): $\frac{3}{4}e_2 - \frac{7}{36}e_2 = \frac{20}{36}e_2 = 5 \quad \Rightarrow \quad e_2 = 9\,\mathrm{V}$

Example 11.4: solve

- (1): $\frac{5}{4}e_1 - \frac{1}{2}e_2 - \frac{1}{2}e_3 = 5$; put $e_1 = e_2 - e_3$ from (3): $\frac{3}{4}e_2 - \frac{7}{4}e_3 = 5$ (4)
- (2): $-e_1 + \frac{5}{6}e_2 + \frac{1}{2}e_3 = 0$; put (3): $-\frac{1}{6}e_2 + \frac{3}{2}e_3 = 0 \Rightarrow e_3 = \frac{1}{9}e_2$ (5)
- (4) and (5): $\frac{3}{4}e_2 - \frac{7}{36}e_2 = \frac{20}{36}e_2 = 5 \quad \Rightarrow \quad e_2 = 9\,\mathrm{V}$
- Then $e_3 = 1\,\mathrm{V}$ and $e_1 = e_2 - e_3 = 8\,\mathrm{V}$.

Example 11.4: solve

- (1): $\frac{5}{4}e_1 - \frac{1}{2}e_2 - \frac{1}{2}e_3 = 5$; put $e_1 = e_2 - e_3$ from (3): $\frac{3}{4}e_2 - \frac{7}{4}e_3 = 5$ (4)
- (2): $-e_1 + \frac{5}{6}e_2 + \frac{1}{2}e_3 = 0$; put (3): $-\frac{1}{6}e_2 + \frac{3}{2}e_3 = 0 \Rightarrow e_3 = \frac{1}{9}e_2$ (5)
- (4) and (5): $\frac{3}{4}e_2 - \frac{7}{36}e_2 = \frac{20}{36}e_2 = 5 \quad \Rightarrow \quad e_2 = 9\,\mathrm{V}$
- Then $e_3 = 1\,\mathrm{V}$ and $e_1 = e_2 - e_3 = 8\,\mathrm{V}$.

$$P_{3\Omega} = \frac{{e_2}^2}{3} = \frac{9^2}{3} = 27\,\mathrm{W}$$

Dependent sources

- A dependent source is handled like an independent one of the same kind.

Dependent sources

- A dependent source is handled like an independent one of the same kind.
- Its control variable (v_x here) is written with node voltages: this gives the constraint equation.

Dependent sources

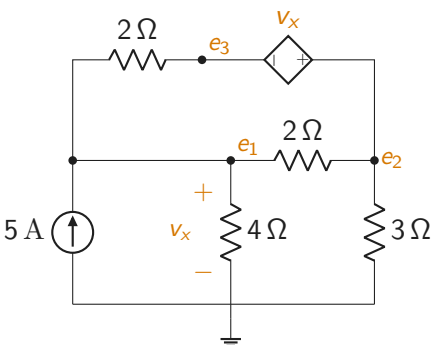
- A dependent source is handled like an independent one of the same kind.
- Its control variable (v_x here) is written with node voltages: this gives the constraint equation.

Remark

If a dependent voltage source is connected between node k and the ground, do not write KCL at node k: its current is not known.

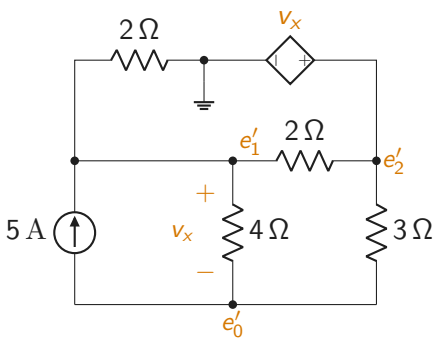
Write the constraint that the dependent source gives instead.

Exercise: choose node 3 as the ground



Question. Redo the example with node 3 as the ground. Does the answer change?

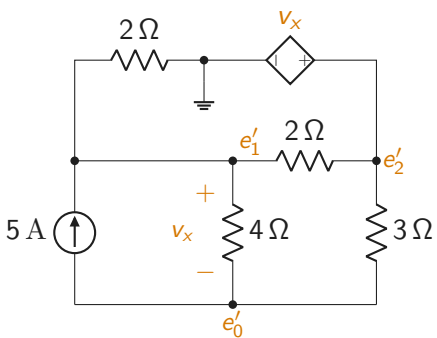
Exercise: choose node 3 as the ground



Question. Redo the example with node 3 as the ground. Does the answer change?

Now v_x is between node 2 and the ground: no supernode is needed.

Exercise: choose node 3 as the ground

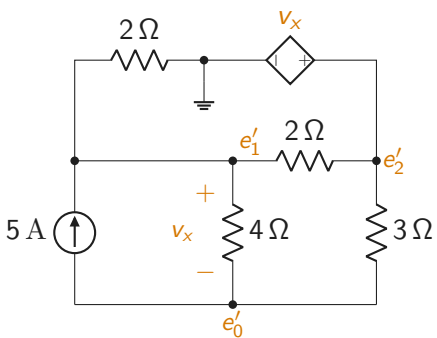


Question. Redo the example with node 3 as the ground. Does the answer change?

Now v_x is between node 2 and the ground: no supernode is needed.

Unknowns: e_1', e_2' and e_0', the old ground.

Exercise: choose node 3 as the ground



Question. Redo the example with node 3 as the ground. Does the answer change?

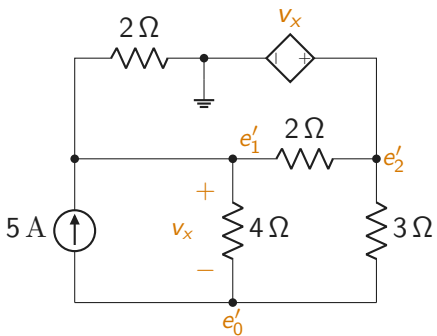
Now v_x is between node 2 and the ground: no supernode is needed.

Unknowns: e_1', e_2' and e_0', the old ground.

Node 1: $$-5 + \frac{e_1'}{2} + \frac{e_1' - e_0'}{4} + \frac{e_1' - e_2'}{2} = 0$$

Node 0: $$5 + \frac{e_0' - e_1'}{4} + \frac{e_0' - e_2'}{3} = 0$$

Exercise: choose node 3 as the ground



Question. Redo the example with node 3 as the ground. Does the answer change?

Now v_x is between node 2 and the ground: no supernode is needed.

Unknowns: e_1', e_2' and e_0', the old ground.

Node 1: $-5 + \dfrac{e_1'}{2} + \dfrac{e_1' - e_0'}{4} + \dfrac{e_1' - e_2'}{2} = 0$

Node 0: $5 + \dfrac{e_0' - e_1'}{4} + \dfrac{e_0' - e_2'}{3} = 0$

Constraint: $e_2' = v_x = e_1' - e_0'$; no KCL at node 2.

Exercise: the solution

- Put the constraint into both KCL equations:

$$3e_1' + e_0' = 20, \qquad -7e_1' + 11e_0' = -60.$$

Exercise: the solution

- Put the constraint into both KCL equations:

$$3e_1' + e_0' = 20, \qquad -7e_1' + 11e_0' = -60.$$

- So $e_1' = 7\,\mathrm{V}$, $e_0' = -1\,\mathrm{V}$, $e_2' = 8\,\mathrm{V}$.

Exercise: the solution

- Put the constraint into both KCL equations:

$$3e_1' + e_0' = 20, \qquad -7e_1' + 11e_0' = -60.$$

- So $e_1' = 7\,\mathrm{V}$, $e_0' = -1\,\mathrm{V}$, $e_2' = 8\,\mathrm{V}$.
- The $3\,\Omega$ is between nodes 2 and 0: $P_{3\Omega} = \dfrac{(e_2' - e_0')^2}{3} = \dfrac{9^2}{3} = 27\,\mathrm{W}$, the same.

Exercise: the solution

- Put the constraint into both KCL equations:

$$3e_1' + e_0' = 20, \qquad -7e_1' + 11e_0' = -60.$$

- So $e_1' = 7\,\mathrm{V}$, $e_0' = -1\,\mathrm{V}$, $e_2' = 8\,\mathrm{V}$.
- The $3\,\Omega$ is between nodes 2 and 0: $P_{3\Omega} = \dfrac{(e_2' - e_0')^2}{3} = \dfrac{9^2}{3} = 27\,\mathrm{W}$, the same.

Every node voltage drops by the same $e_3 = 1\,\mathrm{V}$: $8 \to 7$, $9 \to 8$, $0 \to -1$.

The branch voltages, so the currents and the powers, do not change.

Exercise: the solution

- Put the constraint into both KCL equations:

$$3e_1' + e_0' = 20, \qquad -7e_1' + 11e_0' = -60.$$

- So $e_1' = 7\,\mathrm{V}$, $e_0' = -1\,\mathrm{V}$, $e_2' = 8\,\mathrm{V}$.
- The $3\,\Omega$ is between nodes 2 and 0: $P_{3\Omega} = \dfrac{(e_2' - e_0')^2}{3} = \dfrac{9^2}{3} = 27\,\mathrm{W}$, the same.

Every node voltage drops by the same $e_3 = 1\,\mathrm{V}$: $8 \to 7$, $9 \to 8$, $0 \to -1$.

The branch voltages, so the currents and the powers, do not change.

Tip: put the ground at one end of a voltage source; this saves a supernode.

Summary

In the circuit	What to do
resistors and current sources	KCL at every non-ground node: $Y_n\mathbf{e} = \mathbf{i}_s$, by inspection; Y_n is symmetric
a voltage source from the ground to node k	$e_k = v_s$ is known; no KCL at node k
a voltage source between two nodes	Method I: its current as an unknown, plus the constraint Method II: KCL at the supernode, plus the constraint
a dependent source	write its control variable with node voltages: a constraint
the choice of ground	any node; the branch voltages and currents do not change

After the node voltages: $\mathbf{v} = A^T\mathbf{e}$, then the currents from the element laws.